

RESEARCH ARTICLE

A Hybrid Deep Learning Approach With Explainable AI for Diabetic Retinopathy Classification

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ABSTRACT Diabetic Retinopathy is one of the major cause of vision impairment around the world, yet its impact can be significantly reduced through early detection and timely intervention. Previous screening techniques were dependent on manual detection, which consumed a lot of time and often provided inconsistent results. However, the successful developments in the area of Deep Learning have facilitated the automated and large-scale analysis of the retinal image with better performance. Irrespective of such developments, the use of explainable AI in many studies has received limited attention. This work provides an extensive analysis of several deep learning models like ResNet18, EfficientNetB0, DenseNet121 and MobileNetV2 for multiclass diabetic retinopathy classification, using a unified training framework with SMOTE-based class balancing to address severe data imbalance. In addition to individual models, both hard and soft voting ensemble strategies, as well as a feature-level hybrid architecture combining EfficientNetB0 and DenseNet121, were systematically evaluated. The proposed hybrid model achieved an accuracy of 84.02%, while ensemble learning attained the highest overall accuracy of 85.29%, demonstrating improved robustness and balanced performance across severity levels. Model performance was comprehensively evaluated using accuracy, precision, recall, F1-score, and class-wise ROC-AUC metrics, with per-class AUC values exceeding 0.90 for multiple severity stages. Particularly, model interpretability is introduced with the help of Gradient-weighted Class Activation Mapping (Grad-CAM), where visualization in the form of heatmaps of clinically relevant retinal regions is offered. To move beyond purely qualitative analysis, a quantitative Grad-CAM concentration score was employed, demonstrating more spatially localized and diagnostically focused attention in individual and hybrid model representations. Hence, the results indicate that incorporating complementary convolutional neural network architectures, together with ensemble learning and quantitative explainability, yields a more robust, interpretable, and clinically relevant diabetic retinopathy screening framework.

INDEX TERMS Diabetic retinopathy, deep learning, explainable AI, hybrid model, ensemble learning, model interpretability, gradient-weighted class activation mapping.

I. INTRODUCTION

Diabetic Retinopathy (DR) is a condition generally seen in chronic diabetes patients that affects the patient's eyes

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causing vision loss especially among the working-age population. It develops as a complication of diabetes that occurs due to hyperglycemia, hypertension and elevated cholesterol levels which ultimately causes impairment of the retinal blood vessels, that eventually results in blindness if left untreated. Nearly 75% of individuals living with diabetes for

more than 15 years are likely to develop this condition [1]. DR has been among the leading causes of vision loss over the past few years. Though it remains difficult to be detected during its early stages, few of the symptoms that one must be aware of include gradual blindness and noticeable dark spots in the eyesight. If detected early, treatments like laser therapy and injections can prevent the patients from further damage. In many cases it is seen to initially start with no symptoms and then slowly progress into affecting the patient's vision. However, if detected late then the consequences may be severe. Worldwide, nearly 500 million individuals are affected by diabetes, and approximately one-third of them develop DR [2]. DR generally advances through five distinct stages: No DR, Mild DR, Moderate DR, Severe DR, and Proliferative DR. The last stage causes growth of abnormal blood vessels which eventually leads to bleeding inside the eyes hence having a greater probability of causing permanent blindness. Naturally one can stop the risk of developing DR by keeping the blood sugar, blood pressure and cholesterol level under control. This can often be done by maintaining a healthy lifestyle and a balanced diet.

Before the modern Artificial Intelligence imaging using deep learning was introduced for detection, people made use of a primary method such as visiting the ophthalmologists for a routine eye checkup. This method initially involves dilating the patient's pupil using special eyedrops and further using a device called ophthalmoscope to look into the retina and examine any possible retinal swelling or development of abnormal blood vessels. This technique is time-consuming and requires expensive equipment, hence making it difficult for people to access it in rural areas [3]. Eventually would lead to late detection, causing vision loss in patients.

However, advancements in technology have enabled a significant transformation in DR detection and classification, making the process faster and more preventive. In recent years, deep learning has emerged as a powerful approach in ophthalmology, enabling the analysis of medical images to identify features such as microaneurysms, exudates, and hemorrhages. Further, convolutional neural network (CNN) has shown high accuracy in the medical imaging tasks, including DR detection [4], to automatically grade the severity of DR in patients ranging from 0 to 4, where 0 and 4 corresponds to No DR and Proliferative DR respectively. Therefore, deep learning-based computer assisted fundus imaging systems demonstrate significant potential for accurate detection of DR at the earlier stages [5], hence preventing the risk of vision loss and stands reliable as well as affordable for large-scale screening of patients. Despite the significant progress achieved in deep learning-based DR detection, several practical and technical challenges continue to limit its clinical reliability. Existing studies often struggle with severe class imbalance inherent to retinal datasets, leading to reduced sensitivity in underrepresented severity stages. Moreover, many approaches rely on a single CNN archi-

tecture, which restricts robustness and generalization across diverse clinical conditions. Although ensemble learning has been explored, several studies lack clear technical formulations and systematic comparisons of ensemble strategies such as hard and soft voting strategies, hence reducing methodological transparency. In addition, limited attention has been given to hybrid feature fusion frameworks that explicitly combine complementary architectural strengths for fine-grained multiclass DR grading. A further critical limitation is the insufficient incorporation of explainability mechanisms, as many deep learning models lack transparency in their decision-making processes, restricting clinical trust and real-world adoption. Recent hybrid and explainable deep learning frameworks emphasize the value of combining model-driven regularization with data-driven learning to enhance robustness and interpretability in complex visual analysis tasks [6], [7]. Although originally developed in a different application domain, these works are referenced for their methodological contributions to hybrid modeling and explainable AI, which are conceptually aligned with medical image analysis challenges. Inspired by such methodological advances, the present study adopts a hybrid and ensemble-based framework with explicit interpretability for multiclass DR classification. Further, this research addresses these limitations through a balanced and explainable deep learning framework. The key contributions that this work aims to provide are as follows:

- Multiple pretrained CNNs - ResNet18, EfficientNetB0, DenseNet121, and MobileNetV2 - were systematically trained and evaluated for five-class DR severity classification using fundus images.
- Hard and soft voting ensemble strategies were implemented across the pretrained models to improve prediction consistency and achieve incremental performance gains over individual architectures.
- A custom hybrid CNN architecture combining EfficientNetB0 and DenseNet121 was developed to use the complementary feature representations, resulting in improved classification performance compared to single and ensemble models.
- Model interpretability was enhanced using Grad-CAM quantitative scores and qualitative visualizations, providing clinically meaningful insights by highlighting retinal regions relevant to DR severity predictions.

Collectively, these contributions aim to improve robustness, interpretability, and clinical applicability of deep learning-based DR screening systems.

The rest of this paper is structured as follows: Section II reviews related works, providing an overview of previous research on DR. The third section outlines the Methodology, how the dataset was described, the preprocessing procedures undertaken, the architectures of the models, the ensemble and hybrid strategies. The following section, Section IV presents the experimental setup and training convergence behavior, followed by performance comparison of individual,

ensemble, and hybrid models. Subsequent subsections provide computational complexity analysis, ablation studies, detailed performance evaluation using confusion matrices and ROC-AUC curves, and finally model interpretability analysis using Grad-CAM visualizations.

II. RELATED WORK

There is a considerable amount of literature already on DR classification using various algorithms, methodologies, and procedures. Early deep learning-based studies primarily focused on adapting established CNN architectures for DR classification. Works such as [8] and [9] employed VGG and ResNet-based models on datasets, integrating preprocessing strategies such as contrast enhancement and Histogram equalization (HE) to improve lesion visibility. However, these methods showed limited generalization across severity grades.

Subsequent studies emphasized deeper architectures and optimized training strategies, often incorporating preprocessing to enhance feature representation [10], [11]. Beyond single-backbone optimization, multistage training frameworks incorporating diverse architectures—such as Inception-ResNetV2, VGG variants, DenseNet121, MobileNetV2, and EfficientNet2L—were explored, combining feature extraction with fine-tuning to further enhance classification performance [12]. Recent studies have addressed the inability of conventional deep learning models to capture predictive uncertainty in DR detection by incorporating Bayesian extensions into CNN architectures [13], [14]. These approaches integrate techniques such as Monte Carlo Dropout, variational inference, and deterministic inference within DenseNet-based and generic Bayesian CNN frameworks, enabling probabilistic modeling of prediction confidence. Evaluations conducted on combinations of APTOS-2019, DDR, and Messidor-2 datasets demonstrated improved classification performance alongside more reliable uncertainty estimation, highlighting their potential relevance for clinical decision-making scenarios.

Beyond uncertainty-aware modeling, multitask and lesion-aware learning strategies have been explored to improve DR classification. Multitask learning frameworks that jointly model classification and regression through shared feature representations have demonstrated strong severity grading performance, achieving weighted kappa scores of 0.90 on APTOS and 0.88 on EyePACS [15]. In parallel, to address the limited interpretability of deep learning models, lesion-aware pipelines integrating segmentation and classification have been proposed. These approaches leverage preprocessing and lesion-specific segmentation modules, including U-Net based architectures for exudate and red lesion detection, to generate lesion-focused features for downstream DR severity classification [16], [17]. Hybrid learning frameworks have been widely explored to improve DR classification by combining complementary modeling strategies. These include separating deep feature extraction from classification using ensemble learners such

as Random Forest and XGBoost [18], integrating CNNs with recurrent models for enhanced severity-level prediction [19], and designing multi-stage pipelines that incorporate image enhancement, feature extraction, segmentation, and ensemble-based classification [20]. Architectural fusion approaches that ensemble outputs from multiple CNN backbones have also been proposed to improve class-wise prediction stability across DR severity levels [21]. While effective, such hybrid frameworks often increase system complexity.

While many of the previously mentioned studies have contributed to advancing classification performance through hybrid strategies or architectural modifications, recent work has increasingly emphasized the need for methodological rigor, explainability, and clinically trustworthy evaluation in medical imaging systems. In this direction, the studies reported in [22] and [23] emphasize explainable and hybrid deep learning paradigms that integrate interpretability with rigorous evaluation protocols for image analysis. These works demonstrate how explainability mechanisms, when coupled with carefully designed hybrid architectures, can enhance clinical confidence by ensuring that model predictions are grounded in medically relevant image regions while maintaining robust performance assessment. In particular, the comprehensive review in [24] provides an organized discussion of deep learning methods for DR detection spanning supervised, self-supervised, and transformer-based approaches. It examines binary versus multiclass objectives, lesion-specific methods, multi-task learning, attention mechanisms, generative models, CNN backbones, ensembles, and advanced architectures such as CABNet, MCGS-Net, and Vision Transformer (ViT)-based frameworks. While these approaches report strong performance, many rely on lesion-level annotations, large-scale datasets, multimodal inputs, or complex training pipelines, which can limit their reproducibility and practical deployment in resource-constrained clinical environments. In contrast, our proposed work focuses on an evaluation framework built upon widely adopted CNN backbones, emphasizing the systematic comparison, computational efficiency, and quantitative interpretability rather than the architectural novelty alone.

Unlike the transformer-based or lesion-aware models that often require extensive annotation effort or large external datasets, the proposed approach demonstrates how carefully designed ensemble and hybrid CNN strategies, combined with SMOTE-based balancing, and Grad-CAM quantification, can yield robust and interpretable multiclass DR classification using standard fundus images. This positions the proposed method as a complementary alternative to more complex transformer or multimodal pipelines, offering a balanced trade-off between performance, interpretability, and deployability. The studies summarized in Table 1 were selected based on methodological relevance to multiclass DR classification using deep learning. Priority was given to the works introducing representative architectural strategies that align with the focus of this study. The selected works

TABLE 1. Summary of related work .

Author	Objectives	Models/Architecture	Dataset	Key Findings
S. Akhtar et al. [25]	To develop a custom CNN architecture (RSG-Net) that improves multi-class grading accuracy through advanced preprocessing, data augmentation and architectural regularization techniques.	RSG-Net (custom CNN with pooling, batch normalization, dropout and fully connected layers)	Messidor-1	Achieved 99.36% accuracy (4-class) and 99.37% (binary); 100% sensitivity (binary) and 99.79% specificity (4-class). Demonstrated strong generalization, interpretability and computational efficiency.
C. Sen et al. [26]	To design a high-performance multi-class classifier using a structured pipeline integrating preprocessing, augmentation, transfer learning, and deep learning.	ResNet50 and DenseNet 201/121/169 - combined with preprocessing techniques, data augmentation, and transfer learning for robust classification across five DR stages.	APTOS 2019 Blindness Detection dataset	Achieved 96.95% accuracy and less than 1% false negative rate; outperformed other CNN models like ResNet50 and DenseNet121; model effectively distinguished all DR stages with improved reliability, especially in early detection.
T. Beghriche et al. [27]	To design an efficient multi-level fine-tuned deep learning system capable of differentiating between affected and unaffected fundus images using various backbone models.	DenseNet121, MobileNetV2, and Xception models with multi-level fine-tuning	APTOS and DR Gaussian Filtered 224×224 datasets	Xception achieved 97.95% validation accuracy, outperforming DenseNet121 and MobileNetV2; demonstrated competitive results compared to existing works with consistent generalization across datasets.
M. T. Islam et al. [28]	To demonstrate the viability of retinal images alone for predicting systemic diabetes without clinical biomarkers.	Multi-stage CNN architecture (DiaNet), initialized with DenseNet-121, fine-tuned on EyePACS and QBB data.	EyePACS (for pretraining) and QBB retinal image dataset (Qatari population)	DiaNet achieved 84.47% accuracy, outperforming QBBNet and prior models based on clinical biomarkers; CAM visualizations highlighted biologically relevant retinal regions linked to diabetes and comorbidities.
S. Wan et al. [29]	To explore the classification potential of CNN models on fundus images using transfer learning and hyperparameter tuning.	VGGNet-s, AlexNet, GoogleNet and ResNet with data augmentation.	Fundus-image dataset sourced from Kaggle	VGGNet-s achieved the highest accuracy of 95.68%. Other models underperformed due to overfitting on limited data, highlighting the importance of lightweight architectures

collectively capture the progression of methodological trends relevant to automated DR grading.

III. METHODOLOGY

This section presents the experimental setup employed for classifying DR using deep learning in image analysis. The research design investigated the capabilities of different

architectures of CNN in retrieving the severity level of DR using the retinal fundus images. The workflow consists of steps - dataset description, preprocessing steps, individual and hybrid model architecture, ensemble methodology, training and assessment. The proposed method will test the extent of application of widely used deep learning models to the real-world conditions of class variations. Further, the

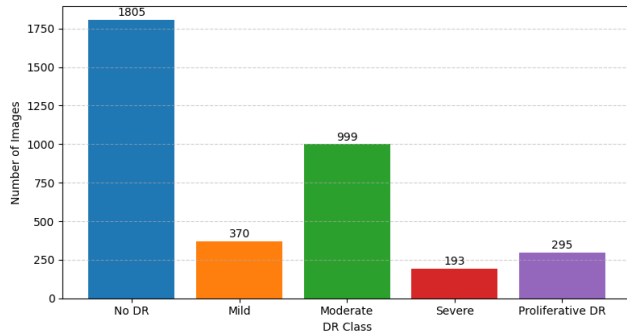


FIGURE 1. Dataset class distribution.

steps of development and assessment process are described in following subsections.

A. DATASET DESCRIPTION

This research primarily utilizes the APTOS 2019 Blindness Detection dataset [30], which is publicly available on the Kaggle platform. The dataset consists of high-resolution color fundus images of the retina, captured under diverse imaging conditions reflecting the real-world clinical variability for the purpose of model development and evaluation. Furthermore, the dataset includes multiple samples across five categories: No DR, Mild DR, Moderate DR, Severe DR, and Proliferative DR.

For this research, 3,662 retinal fundus images, each with a corresponding ground truth label were used. These images were divided into training, validation and internal testing subsets for model development and evaluation. However, the dataset is severely imbalanced as shown in Figure 1, with a disproportionately large number of images labeled as ‘No DR’ compared to other higher severity grades. This existing class imbalance tend to affect model performance, which often leads to biased predictions favoring the majority class and reduced sensitivity in detecting less-represented, yet clinically important DR stages. Consequently, the model can end up classifying with high accuracy in general but will fail to detect images of severe or proliferative DR. Most of the images are of the No DR type, and they constitute the larger part of the dataset with 49%. Moderate DR images comprise the second-largest with approximately 27%, then Mild DR with 10%, Proliferative DR with 8% and Severe DR with approximately 5% of the number of images. This distribution was used consistently for DR classification.

B. DATA PREPROCESSING

Data, as the central unit of deep learning, fall directly under the quality of both input data and data processing, which can influence the accuracy of predictions [31]. Preprocessing is a crucial step in image preparation prior to analysis as it is in medical images where inconsistencies in image size, contrast and quality is usual. In this work, preprocessing was applied to fundus images to evaluate how effectively deep learning models can handle data that reflects real-world

TABLE 2. Image-level distribution across dataset splits.

DR Grade	Training (70%)	Validation (20%)	Test (10%)
Class 0	1263	361	181
Class 1	259	74	37
Class 2	699	200	100
Class 3	135	39	19
Class 4	206	59	30
Total Images	2562	733	367

clinical variability. The architecture diagram of proposed framework is illustrated in Figure 2.

The preprocessing stage started with the preparation of class labels and splitting of dataset to ensure in a given subset the model can get trained from decent number of images. The categorical labels representing the DR severity levels were retained as integer values and used directly for classification. The labeled dataset was divided into training (70%), validation (20%), and internal test (10%) subsets using stratified sampling to preserve the original class distribution across all splits. Given that the APTOS 2019 dataset only offers one fundus picture per eye and lacks patient-level identifiers, each picture was considered an independent example to ensure that there was no image-level overlap between subsets. All splitting operations were performed prior to data augmentation to further prevent data leakage across subsets. It is important to note that patient or subject-level separation could not be enforced, as the APTOS 2019 dataset does not provide any patient identifiers or metadata linking images to the same individual. Each record contains only a single fundus image and its associated DR grade without subject-level traceability. Therefore, images were treated as independent samples, and the split was performed strictly at the image level. This constraint represents an inherent dataset limitation rather than a methodological oversight and is consistent with previously published studies utilizing the APTOS 2019 dataset. Further, the exact image-level distribution across the training, validation, and test subsets prior to augmentation and SMOTE processing is summarized in Table 2, ensuring full transparency of the data partitioning protocol.

The 70/20/10 split was selected to provide sufficient data for model learning while allocating a relatively larger validation set to have stable training monitoring, validation-based model selection, and overfitting control in the presence of severe class imbalance. The internal test set was always kept separate from training and validation to allow an unbiased evaluation of the performance. All images were resized to a uniform resolution of 224×224 pixels to align with the standard input dimensions of CNNs and to reduce computational overhead during training. Pixel values were normalized using a mean of [0.485, 0.456, 0.406] and a standard deviation of [0.229, 0.224, 0.225] values specific to ImageNet, which is used as the pretraining values for transfer learning models used in this work, hence ensuring input image data consistency.

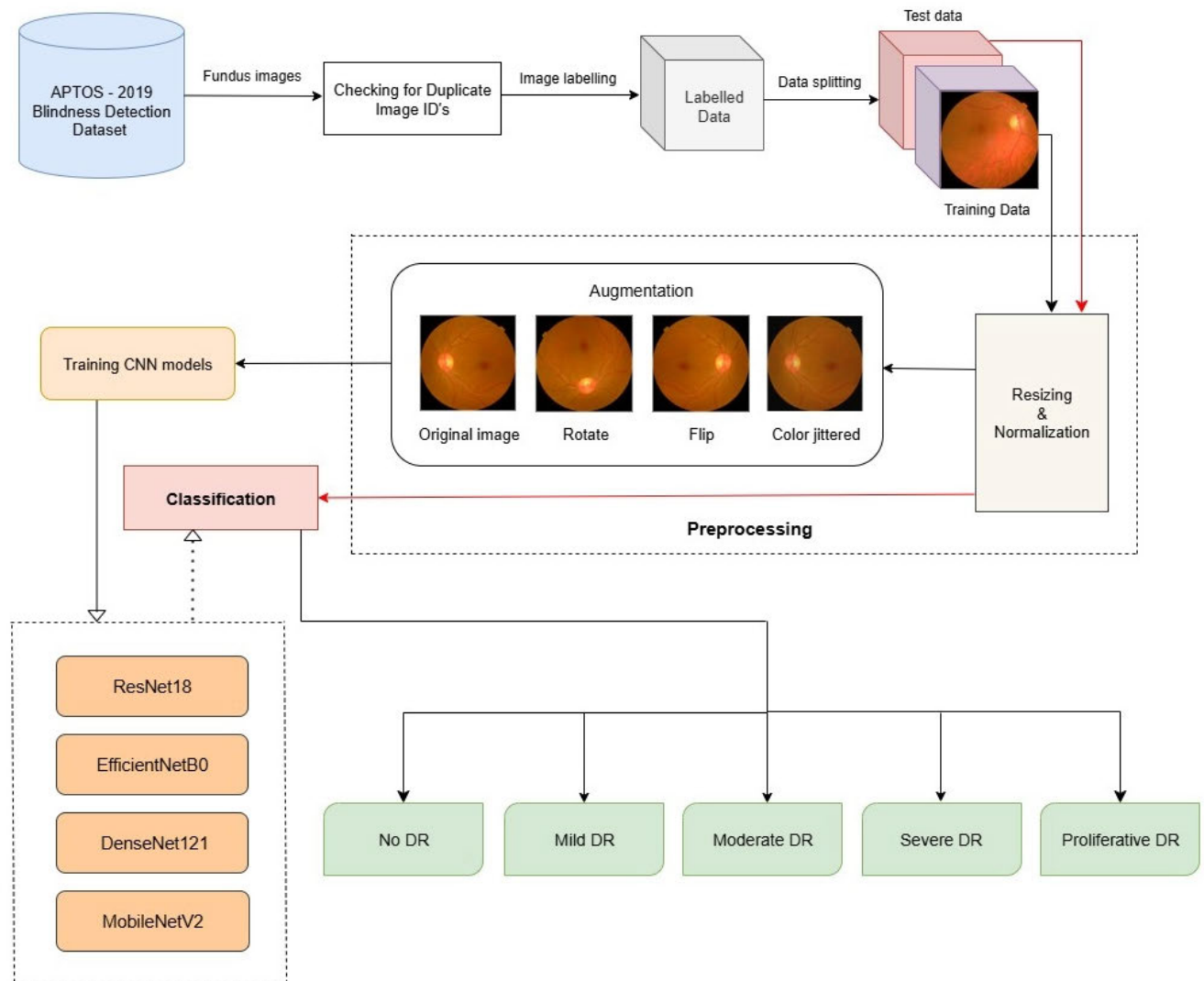


FIGURE 2. Architecture diagram of the proposed deep learning approach for DR classification.

Due to the severe class imbalance present in the APTOS-2019 dataset, particularly for higher DR severity grades, Synthetic Minority Over-sampling Technique (SMOTE) [32] was used to increase the minority class during training. In this study, SMOTE was applied at the image-sample level by operating on flattened, normalized pixel space representations of retinal fundus images within the training and validation subsets only, while the test set remained completely untouched. The resulting class-wise sample counts before and after SMOTE for each data split are summarized in Table 3, ensuring transparency of the balancing process. Although SMOTE was initially designed for tabular data, its controlled use here was not intended to generate clinically interpretable synthetic images but rather to balance the optimization process by exposing the classifier to a more equitable distribution of samples. Other methods for dealing with imbalance, like class-balanced sampling or cost-sensitive loss functions, mainly reweight

the existing samples and do not increase minority-class sample diversity, whereas solely image-level augmentation cannot cope with an extremely skewed severity distribution; thus, the decision was made to use SMOTE to explicitly increase minority-class exposure during learning. In order to prevent the creation of unrealistic artifacts, SMOTE was applied with image-level augmentation strategies, such as rotation, horizontal flipping, and brightness adjustment, and no distortion or instability of minority-class patterns was found. It is important to note that these image-level augmentation operations were applied dynamically during training and did not permanently increase the number of samples in any class. Therefore, the class distribution after augmentation remains identical to the distribution shown in Table 2, and only SMOTE altered the effective training sample counts as shown in Table 3. The selected augmentation techniques simulate realistic acquisition variations such as orientation changes and illumination differences

TABLE 3. Class distribution before and after SMOTE.

Sampling	Split	Class 0	Class 1	Class 2	Class 3	Class 4	Total
Before SMOTE	Train (70%)	1263	259	699	135	206	2562
	Val (20%)	361	74	200	39	59	733
After SMOTE	Train (70%)	1263	1263	1263	1263	1263	6315
	Val (20%)	361	361	361	361	361	1805
No SMOTE	Test (10%)	181	37	100	19	30	367

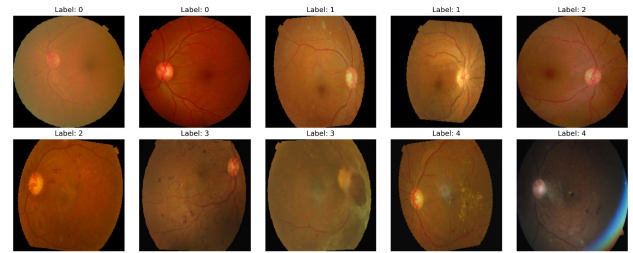
while preserving anatomical consistency, thereby improving generalization without introducing unrealistic distortions. Clinical validity was also supported by stable training-validation convergence, improved minority-class recall, and Grad-CAM explainability results, which showed that the models consistently focused on relevant retinal regions for classification. To further assess the possibility of overfitting to synthetic artifacts introduced by SMOTE, model generalization was evaluated exclusively on the untouched internal test set, where no synthetic samples were present. The absence of abnormal divergence between training and validation curves, together with consistent performance on real test images, indicates that the models learned meaningful disease-related features rather than artificial patterns.

By adopting this strategy, the preprocessing pipeline aims to replicate the real-world scenario where images are processed with limited manual intervention to work with unfiltered clinical data. Moreover, Figure 3 displays the sample fundus images from the dataset. The term “label” on top of each image refers to the severity stage of DR present in each image, which is categorized into five distinct classes: No DR (0), Mild DR (1), Moderate DR (2), Severe DR (3), and Proliferative DR (4), as annotated by the medical experts. These sample images are shown after applying the preprocessing steps which is meant to prepare the data for training while preserving the natural variability of clinical fundus images.

C. MODEL ARCHITECTURE

In this study, the selected CNN architectures were chosen to represent diverse design philosophies and computational characteristics, enabling a comprehensive evaluation of accuracy, efficiency, and deployability in DR classification. Specifically, ResNet18 represents a stable and moderately deep residual architecture, DenseNet121 emphasizes dense feature reuse for fine-grained lesion representation, EfficientNetB0 offers an optimized balance between performance and computational cost through compound scaling, and MobileNetV2 serves as a lightweight model suitable for resource-constrained and real-time screening environments. Together, these architectures span a broad spectrum of model complexity and efficiency, allowing systematic analysis of performance trade-offs and providing a strong foundation for ensemble and hybrid model design.

The compact form of residual network family - ResNet18 [33] introduces residual learning through shortcut

**FIGURE 3.** Retinal Image Samples from Various DR Stages.

connections, which enables the training of deeper networks by mitigating the vanishing gradient problem. It has been selected because of its stability in the analysis of various forms of images. EfficientNetB0 [34] has an efficient compound scaling approach and MBConv blocks that makes this backbone quite efficient in terms of resource requirements, but yet delivers reasonable performance, and is well-suited for fine-grained DR classification tasks on limited GPU resources. One of the closely related networks of the densely connected CNN side of the relation tree - DenseNet121 requires all preceding layers to be feed into the next layer [35]. This helps in feature reuse, which is particularly beneficial in medical image analysis where slight lesion-level details must be retained across layers and is chosen for this study due to its capability to retain fine-grained retinal features using dense connections, which is especially beneficial for differentiating between closely related DR severity levels. Lastly, MobileNetV2 - a lightweight and efficient model designed for mobile devices [36], includes inverted residuals and linear bottlenecks to keep the number of parameters low while maintaining accuracy, making it appropriate for screening environments with limited resources. The combination of these models provides a comprehensive evaluation between accuracy, complexity and suitability in DR detection.

To ensure consistency across all individual pretrained models a uniform training procedure was adopted. Each CNN was initialized using pretrained weights from the ImageNet dataset, enabling effective transfer learning by leveraging rich, generic visual features learned from large-scale natural image data. For all architectures, the original classification layers were replaced with a custom fully connected layer designed to generate predictions for five output classes, corresponding to the five stages of DR severity.

All models process input tensors of shape (batch size $\times$ $3 \times 224 \times 224$), corresponding to RGB fundus images

resized during preprocessing. The internal architectural parameters, including convolutional kernel sizes, follow the original pretrained configurations. Specifically, ResNet18 and DenseNet121 utilize an initial 7×7 convolutional kernel, followed by 3×3 kernels in deeper layers. EfficientNetB0 employs 3×3 and 5×5 kernels within its MBConv blocks to capture multi-scale retinal features, while MobileNetV2 adopts 3×3 depthwise separable convolutions for efficient feature extraction. These configurations were retained without modification to preserve the representational strength of the pretrained models. The stride configuration follows the original pretrained architecture designs. Specifically, stride 2 is applied in the initial convolutional and downsampling layers of the networks to reduce spatial resolution and computational complexity, while stride 1 is used in deeper convolutional layers to preserve fine-grained spatial information critical for retinal lesion detection. Regarding activation functions, all convolutional layers across ResNet18, EfficientNetB0, DenseNet121, and MobileNetV2 employ the Rectified Linear Unit (ReLU) activation function. The final classification layers output raw logits, with the softmax operation implicitly handled by the cross-entropy loss during training.

D. ENSEMBLE METHODOLOGY

Ensemble learning was employed in this study to improve the robustness and generalization performance of DR classification by combining the complementary strengths of multiple CNN architectures. Individual deep learning models often exhibit varying biases toward specific DR severity levels due to differences in architectural design and feature extraction mechanisms. By aggregating predictions from multiple models, ensemble techniques reduce the variance associated with single-model predictions and mitigate the risk of incorrect classification caused by model-specific errors.

In this work, four pretrained CNNs - ResNet18, EfficientNetB0, DenseNet121, and MobileNetV2 - were combined using two widely adopted ensemble strategies: hard voting and soft voting. These approaches differ in how the individual model predictions are aggregated to obtain the final class decision.

1) HARD VOTING ENSEMBLE

In the hard voting approach, each individual model independently predicts a discrete class label for a given input retinal image. The final predicted class is determined by selecting the class that receives the maximum number of votes across all models. All models are treated with equal importance, and confidence information is not considered. Mathematically, the hard voting ensemble prediction is defined in Equation 1.

$$\hat{y}_{hard} = \arg \max_{c \in C} \sum_{m=1}^M I(\hat{y}^{(m)} = c) \quad (1)$$

where M denotes the total number of models in the ensemble, C represents the set of DR severity classes, $\hat{y}^{(m)}$ is the

predicted class label produced by the m -th model, and $I(\cdot)$ is the indicator function.

2) SOFT VOTING ENSEMBLE

Unlike hard voting, the soft voting approach incorporates the confidence of each model's prediction by utilizing class probability estimates. Each model produces a probability distribution over all classes using a softmax activation function, and the probabilities corresponding to the same class are aggregated across models. The final predicted class is the one with the highest combined probability score. The soft voting ensemble prediction is given in Equation 2.

$$\hat{y}_{soft} = \arg \max_{c \in C} \sum_{m=1}^M p^{(m)}(c) \quad (2)$$

where $p^{(m)}(c)$ denotes the predicted probability of class c produced by the m -th model. By aggregating class probability estimates, soft voting provides more stable predictions under model uncertainty.

The ensemble strategies were adopted to enhance classification reliability across all DR severity levels. Since each backbone network captures different visual characteristics of retinal lesions, combining their predictions results in a more comprehensive and stable decision-making process. The performance of both ensemble approaches was quantitatively compared with individual model results to evaluate the effectiveness of model aggregation.

E. HYBRID MODEL ARCHITECTURE

In addition to evaluating individual architectures, this work also introduces a custom hybrid model that combined the feature extraction strengths of EfficientNetB0 and DenseNet121. The motivation behind this fusion lies in using EfficientNetB0's compound scaling efficiency and DenseNet121's dense connectivity to come up with a representation suitable for tasks like DR classification. In deciding to build a hybrid architecture, the main criteria was to achieve a balance between feature representation, which should be complementary, and computationally efficient. EfficientNetB0, through compound scaling and MBConv blocks with depthwise separable convolutions, is highly effective in capturing the structure of both low and mid-level retinal features such as microaneurysms and exudates, thus it is a lightweight model. At the same time, DenseNet121 is based on dense connectivity, which makes it possible to progressively reuse features and keep fine-grained spatial information across layers, thus it is the key in helping to distinguish subtle severity transitions of DR. By leveraging these complementary architectures, the proposed hybrid framework can take advantage of EfficientNetB0's efficient multi-scale feature extraction and DenseNet121's dense semantic representation. The bigger versions such as EfficientNetB4/B7 or DenseNet201 were not chosen because initial studies showed that the increase in performance would be minimal while the costs in terms of computational

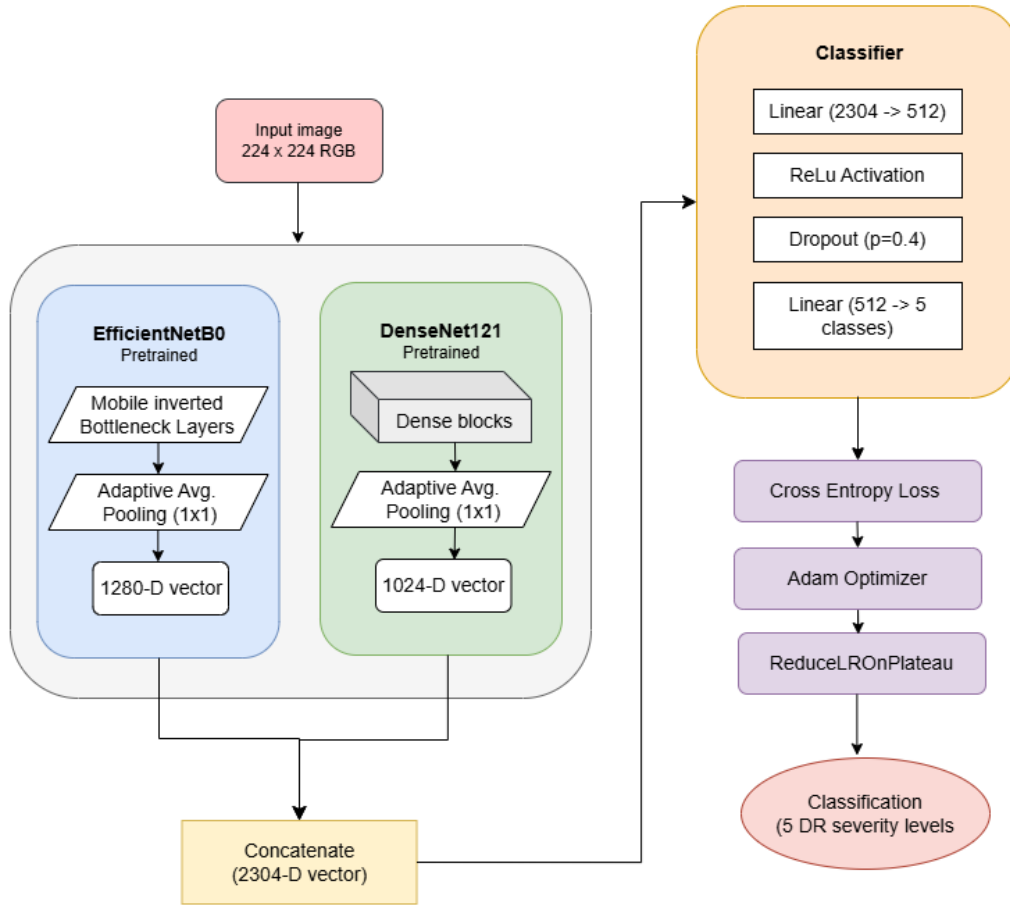


FIGURE 4. Architecture diagram of the proposed hybrid model approach for DR classification.

complexity and overfitting risk on a moderately-sized and imbalanced dataset would be significantly higher. Hence, this combination of two provides an ideal balance between the richness of the representations, the ability to generalize, and practical deployability in clinical screening scenarios. The hybrid architecture includes the convolutional feature extractors of both pre-trained models, the dense block features from DenseNet121 and the mobile inverted bottleneck features from EfficientNetB0 are independently processed through adaptive average pooling with output size 1×1 . These pooled feature maps are in turn flattened to one-dimensional vectors of dimension 1024 (DenseNet121) and 1280 (EfficientNetB0), respectively, and concatenated to give a unified 2304-dimensional feature vector. This combined representation is fed to a fully connected classifier, which comprises two Linear layers (2304 $\rightarrow$ 512) with subsequent ReLU activation, along with a dropout layer with the value for p being 0.4, followed by a linear layer (512 $\rightarrow$ 5) which makes the five DR severity predictions. A structured layer-wise summary of the hybrid architecture, including backbone components, fusion stage, and classifier layers, is provided in Table 4. Further, The model was trained using

the Cross Entropy loss function, as shown in Equation 3.

$$L = -\frac{1}{N} \sum_{n=1}^N \sum_{i=1}^C y_{n,i} \log(\hat{y}_{n,i}) \quad (3)$$

where N is the number of samples, C is the number of classes, $y_{n,i}$ is the ground-truth label, and $\hat{y}_{n,i}$ is the predicted probability. Optimization was performed by including the Adam optimizer, which updates its parameters as defined in Equation 4.

$$\theta_t = \theta_{t-1} - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} \quad (4)$$

where $\hat{m}_t$ and $\hat{v}_t$ are the bias-corrected first and second moment estimates, respectively. The learning rate η was set to 1×10^{-4} , and weight decay was applied with a coefficient of 1×10^{-5} .

To prevent overfitting during hybrid model training, a ReduceLROnPlateau scheduler with a patience of three epochs was employed to adaptively reduce the learning rate when validation performance saturated. Class imbalance handling for the hybrid model followed the preprocessing strategy described earlier, with imbalance mitigation applied

TABLE 4. Layer-wise summary of the proposed hybrid classifier architecture.

Layer / Component	Output Dimension	Function
EfficientNetB0 (pre-trained backbone)	1280-D feature vector	Feature extraction
DenseNet121 (pretrained backbone)	1024-D feature vector	Complementary feature extraction
Adaptive Avg Pooling (both branches)	1280 / 1024	Spatial aggregation
Concatenation	2304	Feature fusion
Linear (2304 $\rightarrow$ 512)	512	Feature integration
ReLU Activation	512	Non-linearity
Dropout ($p = 0.4$)	512	Regularization
Linear (512 $\rightarrow$ 5)	5	DR severity classification

Note: Computational complexity metrics (parameters, FLOPs, and model size) are reported in Section IV-D.

TABLE 5. Training hyperparameters and configuration.

Hyperparameter	Value / Description
Input image size	$224 \times 224 \times 3$
Batch size	32
Optimizer	Adam
Learning rate	1×10^{-4}
Number of epochs	20
Loss function	Categorical Cross-Entropy
Weight decay	1×10^{-5}
Data split ratio	70% Train / 20% Validation / 10% Test
Class imbalance handling	SMOTE (training-validation set)
Data augmentation	Rotation, horizontal flip, brightness adjustment
Activation function	ReLU
Stride configuration	Stride 2 (initial/downsampling), Stride 1 (deeper layers)

prior to training. The overall hybrid architecture is illustrated in Figure 4.

F. TRAINING CONFIGURATION AND EVALUATION PROTOCOL

The complete set of training hyperparameters and configuration details used in this study is summarized in Table 5. The dataset was divided in a 7:2:1 ratio for training, validation, and internal testing, keeping the internal test set completely separate from training. To mitigate class imbalance, SMOTE was applied to the training-validation set following the preprocessing strategy outlined in Section III-B. For all individual pretrained CNN models and the ensemble framework, training was performed using the categorical cross-entropy loss function, which is well suited for multi-class DR classification. Data augmentation strategies were also used during training to promote model generalization in addition to image resizing and normalization. All models

were trained using the Adam optimizer over 20 epochs and learning rate of 0.0001 so as to avoid overfitting. Although minor upward trends were visible in certain training curves toward later epochs, validation accuracy and loss did not show consistent or substantial improvement across models beyond this point. Extended training increased the risk of overfitting without yielding meaningful generalization gains. Therefore, 20 epochs were selected as a balanced convergence point based on validation stabilization rather than continued training loss reduction. The batch size was determined through experiments at 32, thus balancing GPU memory limitations and stable gradient estimation, a practice common in medical image classification tasks. Initial tests with smaller batch sizes showed slower convergence, whereas larger batch sizes did not result in significant performance improvements. Dropout was not deliberately added to the convolutional backbones, as the pretrained architectures are already implicitly regularized through depth and batch normalization. Overfitting was prevented by means of data augmentation and SMOTE-based class balancing. Evaluation metrics including accuracy, precision, recall, and F1-score were employed to assess the performance of the model. Moreover, accuracy curve and loss curve of training and validation sets were used to monitor the progress of training of each model architecture. Each model configuration was trained under identical data splits and hyperparameter settings to ensure fair comparison across architectures. To verify training stability, the experiments were re-executed under the same controlled conditions, and consistent performance values were observed. Since transfer learning with pretrained networks was employed, variability due to random initialization was minimal, indicating stable convergence behavior. Training was conducted for a fixed 20 epochs without employing an automated early stopping mechanism. Each epoch corresponds to one complete pass over the training dataset, with model parameter updates performed per mini-batch during backpropagation. Convergence was monitored using validation loss and accuracy curves to ensure that performance stabilized within the selected epoch range.

G. IMPLEMENTATION PLATFORM

The proposed methodology was implemented using the PyTorch deep learning framework in a Python-based Jupyter Notebook environment. This platform was selected for its flexibility in prototyping, training, and evaluating deep learning models for medical image analysis. Detailed experimental execution settings are presented in the Results and Discussion section.

IV. RESULTS AND DISCUSSION

A. EXPERIMENTAL SETUP AND EVALUATION METRICS

All the experiments were conducted in the interactive Python environment (Jupyter). It was done in Notebook 7.2.2 (using Anaconda distribution). The execution and implementation were conducted on PC running Windows 11 with an

TABLE 6. DR classification results.

Models	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
ResNet-18	85.01	84.90	85.01	84.39
EfficientNet-B0	83.65	83.22	83.65	82.81
DenseNet-121	82.83	82.27	82.83	82.12
MobileNet-V2	83.38	83.04	83.38	82.95
Ensemble (Soft voting)	85.29	84.49	85.29	84.58
Ensemble (Hard voting)	85.29	84.52	85.29	84.16
Hybrid model	84.02	83.12	84.02	82.91

Processor Intel(R) Core(TM) - i7-14700 CPU. A 32.0 GB RAM and an NVIDIA GeForce RTX 3060 GPU (16 GB VRAM) with CUDA 11.1 acceleration. All deep learning models were trained and evaluated using GPU acceleration via PyTorch. The training and validation of the models were done using the steps described in Methodology section, including the preprocessing of the datasets via augmentation and optimization strategies.

This section presents the performance of models for DR classification based on widely adopted metrics in medical image classification. The individual models, ensemble and hybrid model were evaluated using the obtained test accuracy, precision, recall and F1-score. Accuracy measured the proportion of correctly classified samples but, due to class imbalance, it was complemented with Precision, Recall and F1-score. Precision displays percentage of rightly guessed cases by each class DR severity, Recall measures the capability of the model to distinguish between the true cases and the F1-score combines both precision and recall, hence providing a harmonic mean of balanced measure. These metrics were consistently applied across all models to enable a fair performance comparison. The Results section is organized to reflect the progression from training behavior to model performance, robustness analysis, and interpretability assessment.

B. TRAINING AND CONVERGENCE ANALYSIS

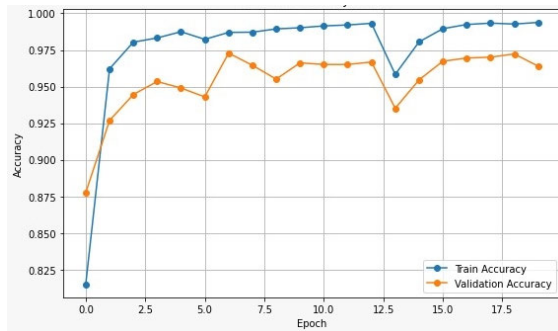
Figure 5 illustrates the training and validation accuracy and loss curves of the four CNN models - ResNet18, EfficientNetB0, DenseNet121, and MobileNetV2 - for DR classification. All models exhibit increasing accuracy and decreasing loss across epochs, indicating effective learning behavior. ResNet18 and DenseNet121 show rapid improvement during the initial epochs, with accuracy curves stabilizing in later stages, suggesting early convergence. In contrast, EfficientNetB0 and MobileNetV2 demonstrate a more gradual and continuous rise in both training and validation accuracy, reflecting improved generalization capability. The close alignment between training and validation curves across all models indicates minimal overfitting. Similarly, the loss curves show a sharp decline during early epochs followed by smooth stabilization, confirming consistent optimization and convergence across all architectures. Extending training to 20 epochs ensured stable convergence across all architectures, with no further performance gains observed beyond this point.

C. PERFORMANCE COMPARISON OF INDIVIDUAL, ENSEMBLE, AND HYBRID MODELS

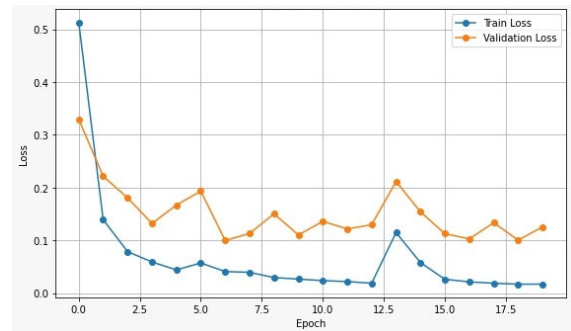
Table 6 presents a comparative performance analysis of individual CNN architectures, ensemble strategies, and the proposed hybrid model for multiclass DR classification. Among the standalone transfer learning models, ResNet-18 achieved the highest accuracy of 85.01%, while EfficientNet-B0, DenseNet-121, and MobileNet-V2 demonstrated competitive performance with accuracies ranging between 82.83% and 83.65%, reflecting comparable discriminative capability across architectures.

The ensemble learning strategies consistently outperformed all individual models. Both soft voting and hard voting ensembles achieved the highest overall accuracy of 85.29%, along with improved precision, recall, and F1-score values, highlighting the advantage of aggregating complementary model predictions to enhance robustness and generalization. The proposed hybrid model, which integrates EfficientNet-B0 and DenseNet-121 through feature-level fusion and SMOTE-based balancing, achieved an accuracy of 84.02%, demonstrating competitive performance relative to individual CNN baselines. Although ResNet-18 achieved slightly higher overall accuracy and F1-score with lower computational cost, the hybrid architecture was designed to investigate complementary feature fusion rather than solely maximize top-line performance. By combining EfficientNet's multi-scale efficiency with DenseNet's dense feature reuse, the hybrid framework enhances representational diversity and supports more localized and stable Grad-CAM attention patterns. Particular attention was given to minority DR severity grades (Severe DR and Proliferative DR), which are underrepresented in the original dataset. The hybrid and ensemble models showed more balanced recall behavior across classes compared to single-backbone architectures, suggesting reduced bias toward the majority No-DR class. From a clinical perspective, robustness across severity levels and interpretability are also critical considerations beyond marginal differences in overall accuracy. To assess whether the observed performance differences between models were consistent across test samples, paired prediction comparisons were examined on the internal test set. The hybrid model demonstrated improved classification outcomes relative to its constituent baseline architectures (DenseNet121 and EfficientNet-B0), indicating that feature fusion contributes positively compared to its individual backbone components. This comparison is therefore presented as a paired performance evaluation supporting the architectural contribution of the hybrid design rather than as a claim of overall superiority across all models.

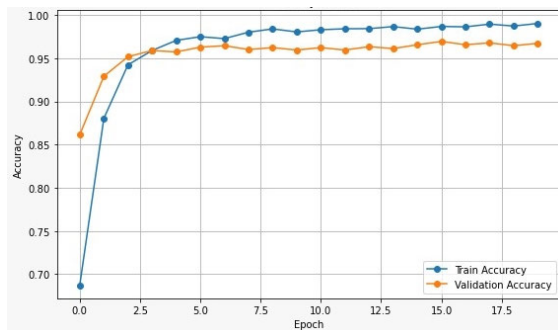
Table 7 presents a contextual comparison of the proposed models with representative state-of-the-art DR classification approaches spanning CNN-based, preprocessing-intensive, traditional machine learning, and transformer-based methods. Earlier CNN models [37] demonstrated the feasibility of automated DR grading, while transfer learning approaches



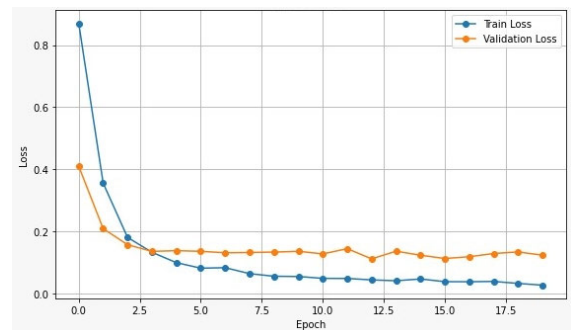
(a) Accuracy Curve (ResNet-18)



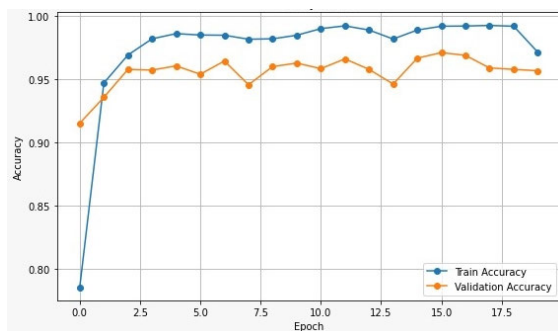
(b) Loss Curve (ResNet-18)



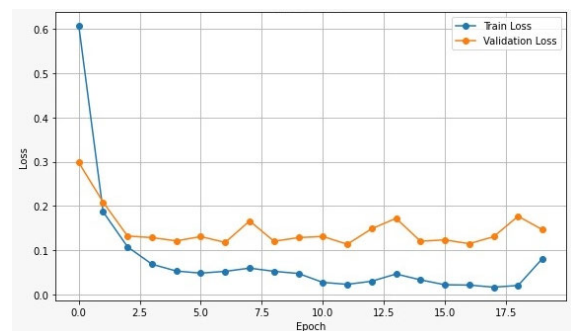
(c) Accuracy Curve (EfficientNet-B0)



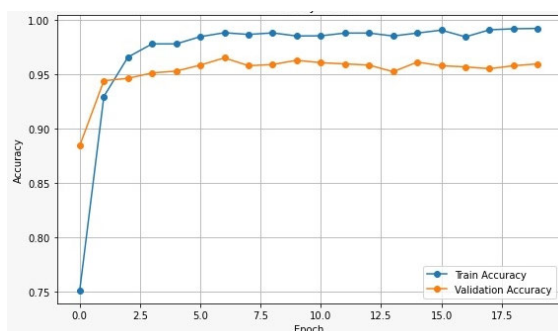
(d) Loss Curve (EfficientNet-B0)



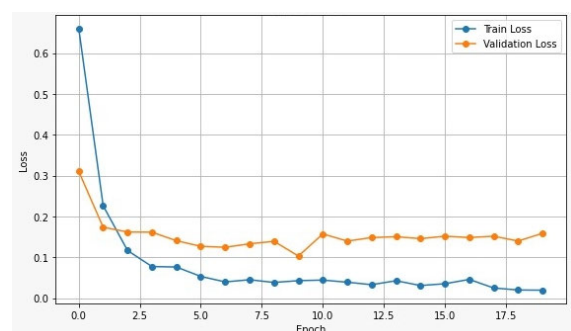
(e) Accuracy Curve (DenseNet-121)



(f) Loss Curve (DenseNet-121)



(g) Accuracy Curve (MobileNet-V2)



(h) Loss Curve (MobileNet-V2)

FIGURE 5. Accuracy and loss curves of four models.

such as the modified Xception model [38] improved performance on APTOS-2019. Methods incorporating extensive preprocessing pipelines, image enhancement or reconstruction strategies such as CLAHE, wavelet transforms, or

GAN-based super-resolution [39], [40], as well as graph-based and uncertainty-aware deep learning frameworks [41], introduce additional modeling stages beyond conventional end-to-end CNN learning. More recent Transformer-based

TABLE 7. Comparison with State-of-the-Art DR classification methods.

Author	Method / Model	Dataset	Accuracy (%)
Pratt et al. [37]	CNN	Kaggle DR Dataset	75.00
Kassani et al. [38]	Modified Xception architecture	APTOS-2019	83.09
Alwakid et al. [39]	DenseNet-121 + CLAHE + ESR-GAN enhancement	APTOS-2019	98.70
Mutawa et al. [40]	CNN-based DR classification with CLAHE and DWT	APTOS + DDR	71.85
Gandor et al. [41]	Random Forest with advanced feature engineering (LBP, GLCM)	APTOS, Aria, DRIVE, E-optha, Eye Diseases Classification (Kaggle), HRF, IDRiD, ODIR, Topcon	80.41
Liu et al. [42]	STMF-DRNet (Swin-TransformerV2, multi-branch attention)	DDR, EyePACS, APTOS-2019, Clinical	77.00
Akter et al. [43]	MobileViT + Graph Convolutional Network with quality & uncertainty modeling	APTOS-2019	98.45
Proposed (Ensemble)	CNN Ensemble (Hard & Soft Voting)	APTOS-2019	85.29
Proposed (Hybrid)	Hybrid CNN (DenseNet121 + EfficientNet-B0)	APTOS-2019	84.02

architectures, such as STMF-DRNet [42], leverage attention mechanisms but typically involve higher computational demands and multi-branch designs. Additionally, graph-enhanced and uncertainty-driven frameworks such as [43] integrate MobileViT feature extraction with Graph Convolutional Networks, quality assessment modules, and uncertainty estimation, reporting very high accuracy on APTOS-2019. While such advanced pipelines can yield strong numerical performance, they involve multi-stage processing, auxiliary learning objectives, and increased training overhead, which may limit reproducibility and real-world clinical deployment.

In contrast, the proposed ensemble and hybrid CNN models achieve competitive accuracies of 85.29% and 84.02%, respectively, while maintaining streamlined end-to-end architectures with integrated explainability. Beyond numerical performance, the proposed study differs from many prior works in three important aspects. First, unlike single-backbone CNN approaches, the hybrid architecture integrates complementary feature representations, enabling improved discrimination of subtle severity transitions in higher DR grades. Second, the inclusion of ensemble learning enhances prediction stability, which is particularly beneficial in imbalanced multiclass medical datasets. Third, the integration of quantitative Grad-CAM analysis addresses a limitation of many prior DR classification studies that focus primarily on accuracy without providing systematic interpretability evidence. The proposed framework therefore aims to balance accuracy, interpretability, and computational feasibility, all of which are critical for real-world DR screening systems.

TABLE 8. Computational Complexity and Deployability Comparison.

Model	Params (M)	GFLOPs	Inference (ms)	Size (MB)
ResNet18	11.18	1.82	3.60	42.73
EfficientNet-B0	4.01	0.41	12.63	15.70
DenseNet121	6.96	2.90	24.85	27.18
MobileNetV2	2.23	0.33	8.49	8.78

D. COMPUTATIONAL COMPLEXITY AND DEPLOYABILITY ANALYSIS

In addition to classification performance, the computational efficiency of the models is a critical factor for practical deployment in clinical environments. Table 8 presents a quantitative comparison of the computational complexity and deployability of the four individual deep learning models evaluated in this study. Among all models, MobileNetV2 is the most computationally efficient, requiring only 2.23 million parameters, 0.33 GFLOPs per image, and 8.49 ms inference time, with a compact model size of 8.78 MB, making it the most suitable architecture for real-time and resource-constrained clinical screening environments. In contrast, DenseNet121 exhibits the highest computational burden in terms of inference time and GFLOPs, requiring 24.85 ms inference time and 2.90 GFLOPs per image, while maintaining 6.96 million parameters and a 27.18 MB model size. Although ResNet18 contains a larger number of parameters (11.18 million) and occupies greater memory (42.73 MB), DenseNet121's dense connectivity pattern results in higher runtime complexity, making it less suitable for low-latency deployment despite its strong feature reuse capability. The proposed hybrid model (DenseNet121 +

TABLE 9. Ablation study of model configurations and performance.

Configuration	DenseNet121	EfficientNetB0	SMOTE	Explicit Dropout	Fusion Strategy	Accuracy (%)	Macro F1-score (%)
EfficientNetB0 only	×	✓	✓	×	–	83.65	82.81
DenseNet121 only	✓	×	✓	×	–	82.83	82.12
Proposed work	✓	✓	✓	✓	Concatenation	84.02	82.91

Note: Explicit Dropout refers to manually inserted dropout layers in the custom classifier head. Backbone-internal regularization remains unchanged.

TABLE 10. Per-Class AUC performance across models.

Model	No DR (0)	Mild (1)	Moderate (2)	Severe (3)	Proliferative DR (4)
ResNet18	0.9928	0.9183	0.9434	0.9044	0.8962
EfficientNet-B0	0.9938	0.9187	0.9361	0.8377	0.8793
DenseNet121	0.9930	0.9241	0.9323	0.8787	0.9037
MobileNetV2	0.9908	0.8903	0.9171	0.8634	0.8504
Hybrid Model	0.9927	0.9100	0.9470	0.8826	0.9535

Note: Per-class AUC values were computed using a one-vs-rest strategy on the internal test set. Higher values indicate stronger class-wise discriminative performance.

TABLE 11. Quantitative Grad-CAM concentration scores across models.

Model	Mean Score	Std. Dev.
ResNet18	0.4498	0.0758
EfficientNet-B0	0.5566	0.0797
DenseNet121	0.3882	0.0749
MobileNetV2	0.4062	0.0817
Hybrid – EfficientNet Branch	0.6586	0.0839
Hybrid – DenseNet Branch	0.4244	0.0868

Note: Grad-CAM concentration scores were computed over 50 test images. Higher values indicate more spatially localized and diagnostically focused activation.

EfficientNet-B0) achieves the highest diagnostic performance by fusing complementary feature representations; however, this comes at the cost of increased computational complexity (14.45 million parameters, 19.86 GFLOPs, and 56.39 MB model size). Overall, these results highlight a clear trade-off between computational efficiency and diagnostic accuracy, with MobileNetV2 being most suitable for deployment-oriented settings and the hybrid model being optimized for accuracy-driven clinical analysis. Despite the higher computational demand of the hybrid architecture, its complexity remains compatible with modern GPU-enabled clinical screening systems, where inference is typically performed in batch mode rather than under strict real-time constraints. The model operates on standardized 224×224 inputs without multi-stage preprocessing, supporting efficient throughput in centralized hospital or cloud-assisted diagnostic workflows. However, deployment on resource-constrained edge devices may require additional optimization strategies such as pruning, quantization, or lightweight model distillation to reduce latency and memory footprint. Thus, the hybrid framework is suitable for accuracy-driven centralized screening scenarios, while lightweight models such as MobileNetV2 remain preferable for edge-based or real-time applications. Although explicit inference latency of the hybrid model was not independently benchmarked, its computational profile relative to the individual backbones indicates suitability for GPU-based batch

screening environments rather than strict real-time processing constraints. The ensemble model does not constitute a standalone network and therefore inherits the computational cost of its constituent models. Similarly, the hybrid architecture reflects the combined parameter count and FLOPs of its backbones, while separate latency benchmarking was not performed.

E. ABLATION STUDY

An ablation study was conducted to analyze the contribution of individual backbone networks and their feature-level fusion within the proposed framework. Table 9 compares the performance of DenseNet121 alone, EfficientNetB0 alone, and their hybrid combination, all trained under identical pre-processing, imbalance handling, and optimization settings. The individual backbones exhibit comparable performance, indicating similar representational capacity when used independently. In contrast, the hybrid model that concatenates features from both networks achieves a clear improvement in both accuracy and macro F1-score. Since the training protocol and imbalance handling strategy remain fixed across configurations, the observed performance gain can be attributed to the complementary nature of EfficientNetB0's multi-scale feature extraction and DenseNet121's dense semantic representations rather than to parameter count or data resampling effects alone. These results confirm that feature-level fusion enables more effective utilization of heterogeneous representations for fine-grained multiclass DR classification.

F. PERFORMANCE ANALYSIS USING CONFUSION MATRIX AND ROC-AUC

In this multiclass DR classification task, the confusion matrix is interpreted using a one-vs-rest strategy for each severity class. For a given class, a True positive (TP) corresponds to retinal fundus images correctly classified into that class. A False positive (FP) represents images incorrectly predicted as belonging to that class while originating from other classes. A False negative (FN) denotes images belonging to that

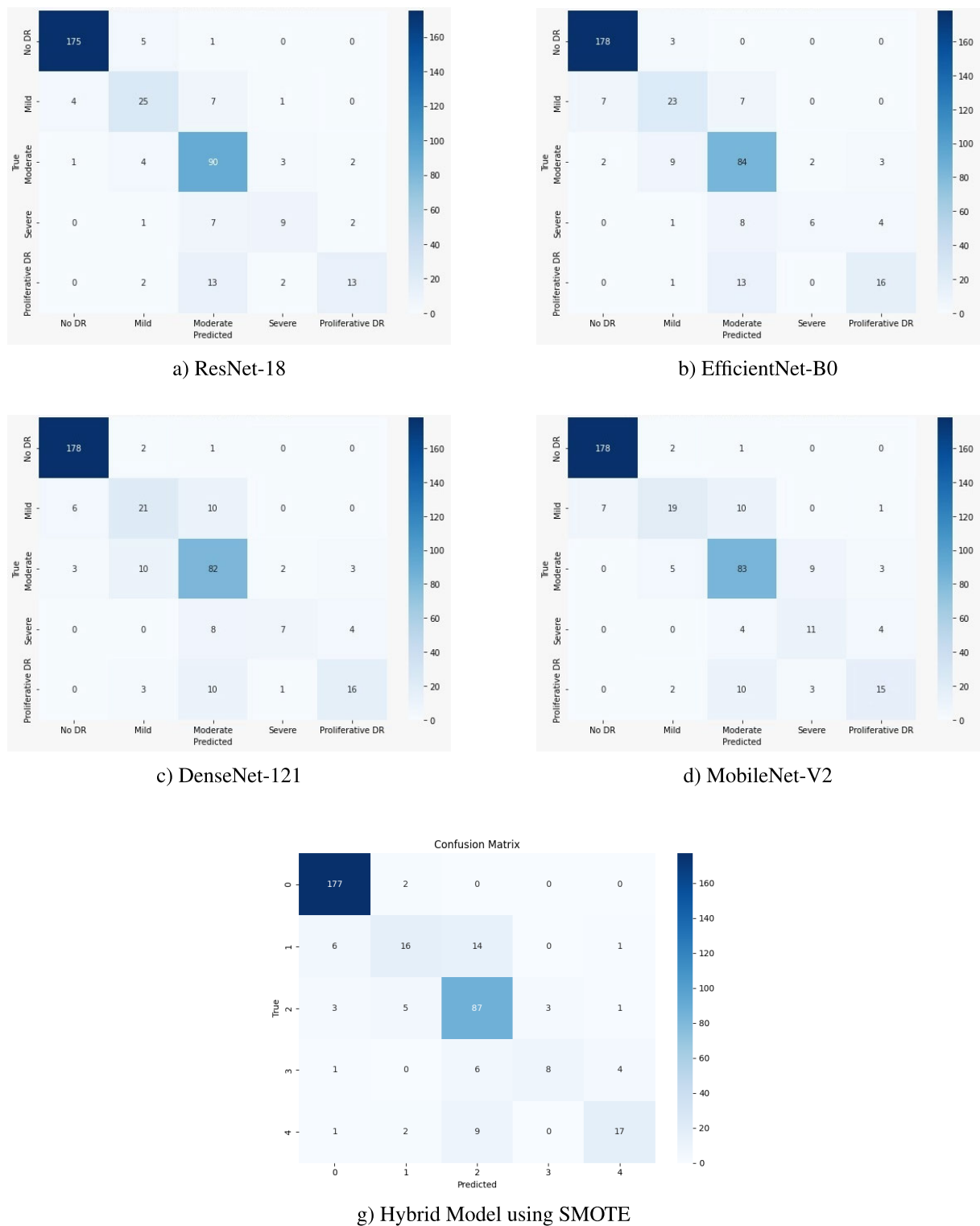
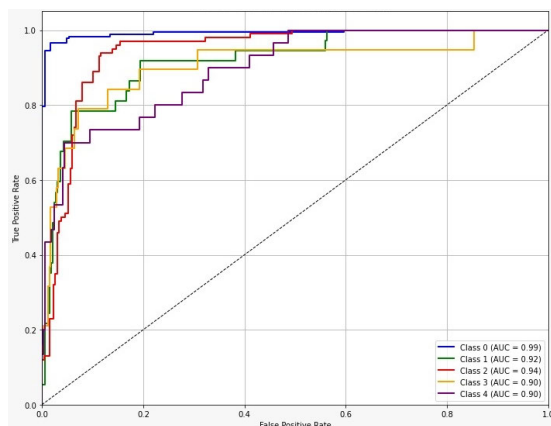


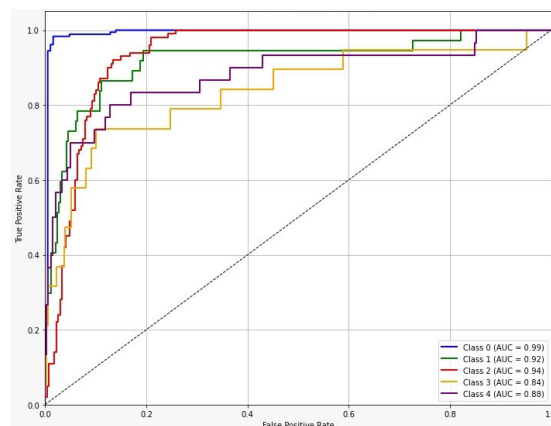
FIGURE 6. Confusion Matrices of Four Deep Learning Models and the Hybrid Model Using SMOTE.

class but misclassified into another severity level. A True negative (TN) corresponds to images correctly classified into all remaining classes. These definitions are consistently used for computing precision, recall, F1-score, and for analyzing the class-wise performance of the proposed models. The

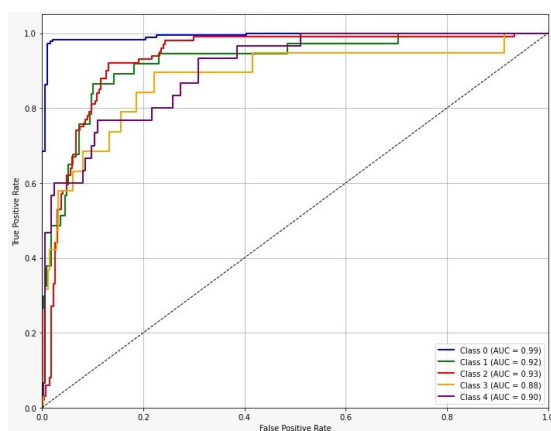
confusion matrices obtained for the various models are as shown in Figure 6, which illustrates the spread of correct and incorrect predictions over five DR severity levels. In all cases, the visible diagonal dominance shows that the models tend to do well in classifying the correct class for most



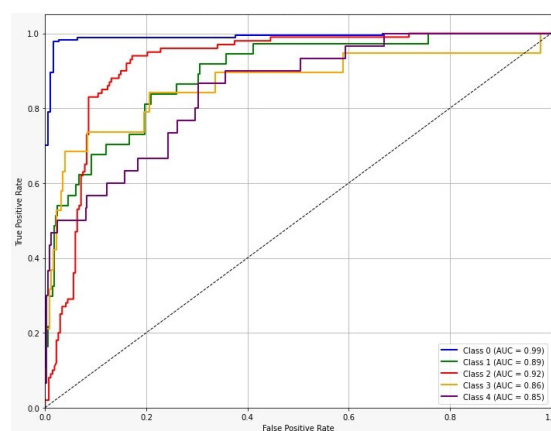
a) ResNet-18



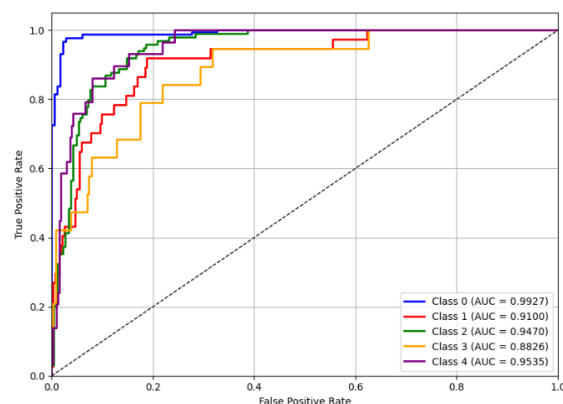
b) EfficientNet-B0



c) DenseNet-121



d) MobileNet-V2

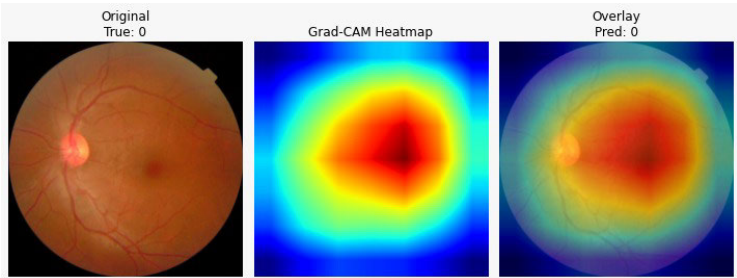


g) Hybrid Model using SMOTE

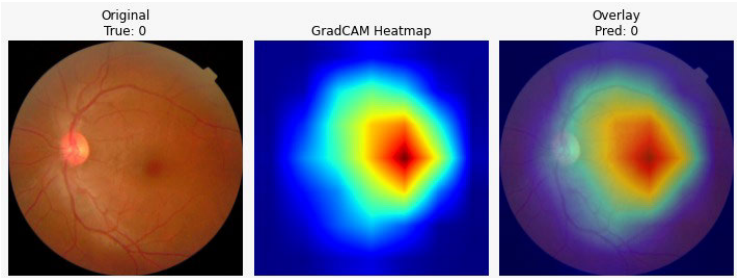
FIGURE 7. AUC Curves of Four Deep Learning Models and the Hybrid Model Using SMOTE.

samples. As expected, the classes with higher representation in the dataset show more consistent recognition, while comparatively lower-frequency classes show some overlap with neighboring classes. Despite these variations, the overall patterns tell that each model is able to differentiate well between the severity levels of DR, thereby supporting their

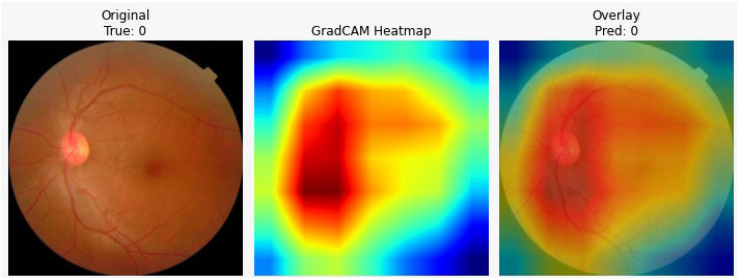
dependability in DR grading tasks. Comparative AUC-ROC (Area under the receiver operating characteristic) curves of all models considered is given in Figure 7, which graphically compares the discriminative performance at class level. This provides total analysis of model performance by determination of the sensitivity and specificity across



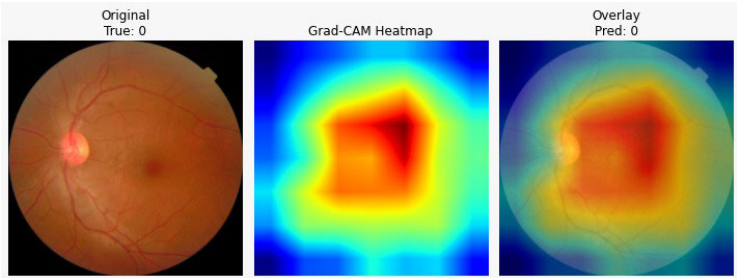
(a) Grad-CAM visualization for ResNet-18



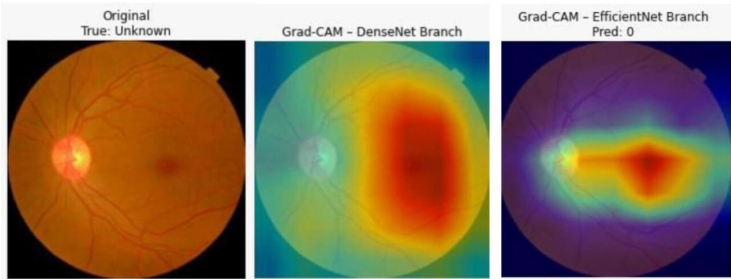
(b) Grad-CAM visualization for EfficientNet-B0



(c) Grad-CAM visualization for DenseNet-121



(d) Grad-CAM visualization for MobileNet-V2



(e) Grad-CAM visualization for Hybrid Model using SMOTE

FIGURE 8. Grad-CAM Visualizations Illustrating Model Attention Regions Across DR Severity Levels.

levels. Compared to a single-value measure, AUC-ROC is a graphical representation of the trade-offs among the FP rate and the TP rate, and presents a comprehensive picture of the ability of each of the models to differentiate between the levels of DR. To complement the visual comparison provided by the ROC-AUC curves, the per-class AUC values for each model are summarized in Table 10. Reporting class-wise AUC scores enables a more granular assessment of discriminative performance across individual DR severity levels, particularly for minority classes where accuracy alone may be misleading. This results indicate that while most models achieve strong discrimination for the No DR and Moderate DR classes, variability is observed in higher severity grades, reflecting the inherent class imbalance and inter-class similarity.

G. MODEL INTERPRETABILITY USING GRAD-CAM

Grad-CAM was employed to enhance the interpretability of the trained models by providing insights into their decision-making process. Grad-CAM generates class-weighted activation maps that highlight regions of retinal fundus images contributing most strongly to a model's prediction, enabling verification of whether the learned representations focus on clinically relevant structures. For the standalone CNN architectures (ResNet18, EfficientNet-B0, DenseNet121, and MobileNetV2), the final convolutional block of each network was selected as the target layer for gradient computation. Each test image was passed through the network, and gradients corresponding to the predicted class were combined with the target feature maps to produce localization maps. These maps were normalized, resized to the input resolution, and overlaid on the original fundus images to visualize salient retinal regions influencing classification decisions.

In the proposed hybrid model, Grad-CAM analysis was performed separately for each branch - namely, the EfficientNet-B0 branch and the DenseNet121 branch - using their respective final convolutional layers. This branch-wise analysis provides clearer insight into how each backbone contributes to the fused decision-making process. Representative Grad-CAM visualizations are shown in Figure 8. It is important to note that although the convolutional backbones were initialized with pretrained weights and only the classifier layers were modified, Grad-CAM localization is influenced by gradients propagated from the classifier to the final convolutional feature maps. Since the hybrid model introduces a feature-fusion mechanism and a distinct classifier head, the gradient signals used to compute Grad-CAM weights differ from those in standalone models. Consequently, variations in attention localization arise from how the fused representation is utilized by the classifier rather than from changes in convolutional filters themselves. To complement the qualitative analysis, a quantitative Grad-CAM concentration score was computed to assess the spatial localization of model attention. Table 11 reports the mean and standard deviation of concentration scores

calculated over 50 test images. EfficientNet-B0 achieved the highest concentration score among individual models, indicating more localized attention on diagnostically relevant retinal regions, whereas DenseNet121 exhibited comparatively more diffused activation patterns. MobileNetV2 and ResNet18 demonstrated moderate localization behavior. In the hybrid architecture, branch-wise analysis reveals that the EfficientNet-B0 branch contributes substantially stronger spatial focus than the DenseNet121 branch, validating the complementary role of feature fusion. These quantitative results substantiate the interpretability claims and confirm that the proposed models consistently attend to clinically meaningful retinal structures.

V. CONCLUSION

This study investigated the effectiveness of multiple deep learning strategies for automated multiclass DR classification, with a particular emphasis on model interpretability, robustness, and clinical relevance. Four widely used CNN architectures - ResNet18, EfficientNetB0, DenseNet121, and MobileNetV2 - were systematically evaluated alongside ensemble learning strategies and a feature-level hybrid architecture combining EfficientNetB0 and DenseNet121. Beyond conventional accuracy-based evaluation, model performance was assessed using precision, recall, F1-score, class-wise ROC-AUC, confusion matrix analysis, and computational complexity analysis. Ensemble learning consistently achieved the highest predictive performance, while the proposed hybrid model demonstrated statistically significant improvements over its constituent architectures. Importantly, quantitative Grad-CAM concentration analysis revealed that EfficientNet-based representations exhibited more spatially localized and diagnostically focused attention, strengthening the interpretability claims beyond qualitative visualization alone. The contribution of this work lies in demonstrating that combining complementary CNN architectures - either through ensemble voting or hybrid feature fusion - can enhance diagnostic reliability while maintaining interpretability through explainable AI techniques. The integration of SMOTE-based balancing and quantitative explainability metrics provides a comprehensive and clinically aligned evaluation framework for DR grading systems.

Despite these encouraging results, this study is limited by the use of a single dataset (APTOS 2019), which may not fully capture variability arising from different imaging devices, acquisition protocols, or population demographics. Although preprocessing normalization and data augmentation were used to mitigate variations in illumination and image acquisition conditions, cross-dataset validation is still necessary to ensure robustness against domain shifts encountered in real clinical environments. Such domain shifts are known to impact DR model generalization and must be explicitly addressed prior to clinical translation. Therefore, future work will focus on extensive multi-center external validation using independent datasets acquired across diverse imaging devices, acquisition protocols, and

patient populations to rigorously assess model generalizability and robustness. Additional directions include exploring alternative loss functions, transformer-based architectures, and integration of multimodal clinical metadata to further improve reliability and clinical applicability. Furthermore, uncertainty-aware prediction and confidence estimation will be investigated to support safer clinical decision-making and real-world deployment in large-scale DR screening programs. These efforts aim to bridge the gap between algorithmic performance and real-world deployment in large-scale DR screening programs.

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